

```

module traffic_light_controller (
    input wire clk,
    input wire rst_n,
    input wire timer_done,
    output reg red,
    output reg grn,
    output reg yel
);
    localparam RED = 2'd0,
               GRN = 2'd1,
               YEL = 2'd2;
    reg [1:0] state, next_state;

    always @(posedge clk) begin
        if (!rst_n)
            state <= RED;
        else
            state <= next_state;
    end

    always @(*) begin
        case (state)
            RED: next_state = timer_done ? GRN : RED;
            GRN: next_state = timer_done ? YEL : GRN;
            YEL: next_state = timer_done ? RED : YEL;
            default: next_state = RED;
        endcase
    end
end

```

```
always @(*) begin
    {red, grn, yel} = 3'b000;

    case (state)
        RED: red = 1'b1;
        GRN: grn = 1'b1;
        YEL: yel = 1'b1;
        default: begin
            red = 1'b0;
            grn = 1'b0;
            yel = 1'b0;
        end
    endcase
end
endmodule
```